

Project Title: AI-Assisted Predictive Maintenance Framework for Coal Handling and Mining Equipment

Host Institution: CSIR-CIMFR [DEMO DATA]

Principal Investigator: Dr. Ananya Rao

Research Domain: Automation & Robotics in Mining

1. Problem Statement

Unexpected mechanical failure of heavy coal handling machinery, conveyor belt gearboxes, and continuous miners causes severe production downtime and safety hazards in underground and opencast coal mines.

2. Project Objectives

- 1. Deploy IoT vibration, thermal, and acoustic telemetry sensors across coal handling equipment.
- 2. Train edge AI predictive maintenance models to forecast component failures 72 hours in advance.
- 3. Deliver an automated maintenance alert dashboard for CIL mine engineers.

3. Technology & Infrastructure

Tri-axial vibration transducers, infrared thermal sensors, ATEX Zone 0 edge computing nodes, PyTorch ML inference models, and wireless mesh communication gateways.

4. Proposed Methodology

Multi-sensor data acquisition from coal handling plant drives, feature extraction, continuous anomaly detection, and field trial validation at CSIR-CIMFR test rig and operating mine sites.

5. Experimental Validation Plan

12-month pilot testing across 3 coal handling conveyors at CSIR-CIMFR facility followed by field deployment at an operating CIL underground coal mine.

6. Expected Outcomes & Deliverables

- 1. Real-time predictive maintenance software suite for coal handling equipment.
- 2. Reduction in unscheduled downtime by 35%.
- 3. Field test report and operational manual.

7. Project Budget & Financial Breakdown

Total Requested Budget: Rs. 48.50 Lakhs

Cost Head Item	Proposed Amount
Equipment and sensor interfaces	Rs. 18.00 Lakhs
Project personnel	Rs. 12.00 Lakhs
Software and computing	Rs. 6.50 Lakhs
Field trials and travel	Rs. 7.00 Lakhs
Contingency	Rs. 3.00 Lakhs

8. Risk Analysis & Mitigation

Harsh underground environmental conditions (dust, moisture) may impair sensor longevity. Mitigation: IP67 ATEX-certified enclosures and redundant sensor channels.

9. Team & Institutional Capability

Principal Investigator: Dr. Ananya Rao, Senior Principal Scientist, CSIR-CIMFR Dhanbad.

10. References

1. CIL Mining Equipment Safety Guidelines (2025).